Supplementing credit risk models

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October 2026

1 Introduction

Accurately predicting the rate of default given empirical evidence is the central challenge of credit risk analysis, and is a necessary undertaking for a bank wanting to make profitable loans. Especially in the age of machine learning, risk modelers look to solve this problem with large amounts of data, some of which may be supplied from third-party vendors such as credit bureaus. However, this third party data is not free, and may not always be available since it relies on external parties; naturally, a bank considering using this information would like to have some quantitative understanding of its usefulness.

In this project, we will examine the effect of credit bureau data on model performance in the context of European consumer finance. Our data set will be taken from the Kaggle competition "Home Credit - Credit Risk Model Stability" which is based on historical data from the Czech[1] lender Home Credit around the time of the COVID-19 pandemic, see figure 1. The full data set contains 1,526,659 cases with 47,994 defaults between the dates of January 1st, 2019 and October 5th, 2020.

2 The models

2.1 Logistic regression

The historical standard for classification models of all shapes and sizes is logistic regression, which is inherently interpretable in the sense that the magnitude and direction of a variable's effect on the classification can be read off from the fit coefficients. In the field of credit risk analysis, this method is nearly always supplemented by the so-called weight of evidence (WOE) transform: for a discrete attribute with k levels, let n and D be the total number of observations and defaults in the data set, with n_j and D_j the number of observations and

[1] It's important to keep in mind that European and American lenders are subject to different legal requirements, and therefore the resulting analysis might not be directly admissible in an American context.

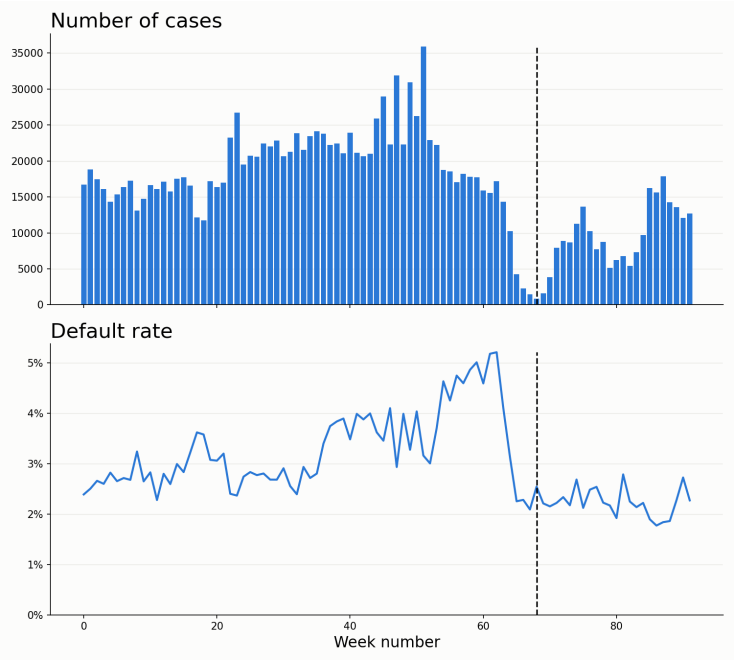


Figure 1: Number of cases (top) and rate of default (bottom) versus week number for the Home Credit data set. Week 0 corresponds to the week of January 3rd, 2019; [check: the text says the data starts January 1st, 2019] week 68 (indicated by the dashed vertical line) corresponds to the week of April 21st, 2020.

defaults associated with the j-th level of the attribute. Then

$$\mathrm{WOE}_j w = \log\left(\frac{D(n_j - D_j)}{D_j(n - D)}\right) \tag{1}$$

is the weight of evidence for level j. Note that WOE is non-linear, but still preserves interpretability since the contribution of each level is explicitly given. We note that a count of 0.5 is typically added to empty levels to keep the WOE finite. Furthermore, feature selection is typically performed via the information value

$$\mathrm{IV} = \sum_{j=1}^{k}\left(\frac{n_j - D_j}{n - D} - \frac{D_j}{D}\right)\log\left(\frac{D(n_j - D_j)}{D_j(n - D)}\right) \tag{2}$$

which roughly speaking captures the degree of dependence of the target as a function of a given variable. In this work, we employ the rule-of-thumb from [1] that features are considered predictive only if their IV exceeds 0.02. Note that we have assumed all features have discrete levels so that using the WOE transform requires binning of continuous variables, the precise number and widths of which can be optimized, though here we will use ten decile bins for convenience. Furthermore, categorical levels with less than 1% of the training cases are grouped together, and missing values form their own group.

2.2 Gradient boosting

Boosting is a machine learning technique where an ensemble of decision trees is constructed via gradient descent iteratively applied to the data set. Empirically, boosted models perform very well on large data sets, and are the default weapon-of-choice for many competition-style hackathons à la Kaggle. Many standard packages exist for implementing gradient boosted methods: here we will employ CatBoost, since our data set contains many categorical features, and since CatBoost tends to perform well out-of-the-box. Our model is fitted with 300 iterations, a learning rate of 0.05 and depth 6. No early stopping is used. Like most ML models, CatBoost can benefit greatly from hyperparameter tuning, though we keep this to a minimum here since our goal is not to produce the strongest model possible.

Gradient boosted models frequently outperform simple regressions, though they can be prone to overfitting, and are not easy to interpret internally, e.g. at the level of the decision tree, as logistic regression models are. Feature selection is also less important for boosted models, as the decision tree will quickly learn which features are important; nevertheless, for the sake of a fair comparison we will feed the same features into both models, though we should remark that this will cause CatBoost to under-perform.

3 The role of credit bureau data

In order to evaluate the importance of external credit bureau data, we will train both a logistic regression and a gradient boosted model via CatBoost with and

AUC	Cases	LR	LR+	CB	CB+
train, pre-covid	1300320	0.773	0.809	0.795	0.827
train, post-covid	94791	0.802	0.835	0.819	0.849
test	131548	0.811	0.842	0.830	0.856

Table 1: Model performance by AUC during training pre- and post-COVID lockdown. The sign + in the column name indicates the inclusion of credit bureau data. In-sample values are included to demonstrate the impact of the COVID lockdown.

without access to credit bureau data. To make the comparison consistent, we will perform feature selection by IV and corelation for both logistic regression and CatBoost. To avoid multicollinearity, we will eliminate features with correlation greater than 0.95 using IV to decide which features to keep, leaving 96 of 214 attributes without the bureau data and 204 of 710 attributes with it. We also drop external credit scores provided by the credit bureaus; the rational for this decision is that including these metrics dominates the signal in our simple analysis, so that we gain more insight by excluding them. All models are trained on weeks 0 to 80 (1,395,111 cases, default rate 0.0324) and tested on weeks 81 to 91 (131,548 cases, default rate 0.0212).

We find that including the external credit bureau data improves the model performance as measured by area under the curve (AUC) by 0.031 for logistic regression and 0.026 for CatBoost on the test weeks; full week-by-week results are presented in figure 2 and summarized in table 1. CatBoost leads logistic regression by 0.019 AUC without the bureau data and by 0.014 with it, but logistic regression with the bureau data still beats CatBoost without it by 0.013.

In table 2 we further compare the calibration, i.e. predicted and observed rates of default, for the training period. Unsurprisingly, the COVID-19 lockdown has a significant negative impact on model calibration, with all four models over-predicting the rate of default by almost a full percentage point. Much of this decrease in the rate of default was likely caused by default moratoriums imposed by various regulatory agencies, which lasted well past the final week in the data set. In this context, it is reasonable to assume that a bank would want to recalibrate its PD distributions as quickly as possible; for the sake of our analysis, we will recalibrate to the observed rate of .0230 in the period from week 68 to 80, i.e. the part of the training period which takes place after the start of lockdown. Each model's intercept is shifted, i.e. the same change in log odds is applied to every case, so that its mean PD over weeks 68 to 80 equals the observed rate. After recalibration the mean predicted PD in the test weeks is .0229 to .0241 against an observed .0212 (last row of table 2). This result has no effect on AUC metrics, which only rely on the relative ranking of individual accounts, and so are not affected by a global shift in PD.

We can make a determination on the value provided by the credit bureau data by computing the break-even cost under reasonable assumptions. We em-

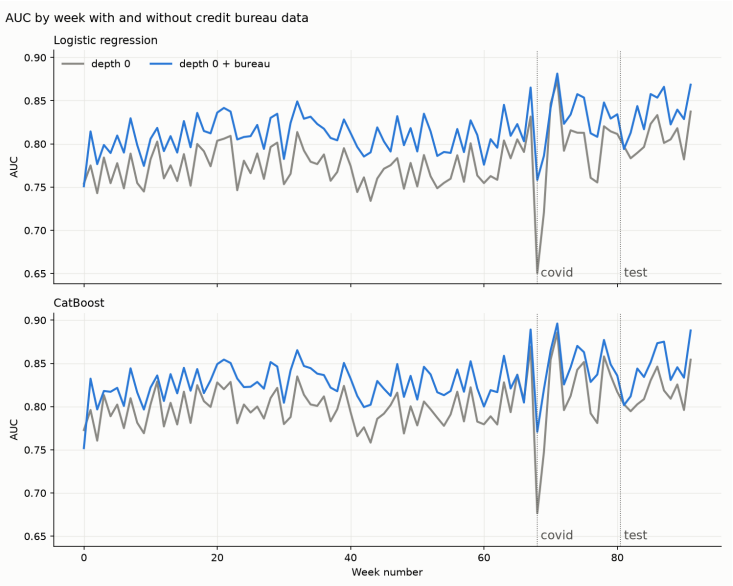


Figure 2: Weekly AUC scores for the logistic regression and CatBoost models with and without external credit bureau data. All models are trained on data from the first 80 weeks, including the initial lockdown period of the COVID-19 pandemic and tested on subsequent weeks. AUCs up to week 80 are in-sample. The dip around week 68 comes from weeks with only 1,000 to 2,000 cases, so it can plausibly be attributed to noise.

Calibration	True default rate	LR	LR+	CB	CB+
train, pre-covid	.0331	.0327	.0327	.0326	.0326
train, post-covid	.0230	.0296	.0276	.0283	.0275
test	.0212	.0295	.0283	.0288	.0288
test, recalibrated	.0212	.0229	.0236	.0234	.0241

Table 2: Predicted and observed rates of default in training data pre- and post-week 68, and for test data. Predicted default rates are mean values computed from the predicted PDs. The sign + in the column name indicates the inclusion of credit bureau data. The last row uses the PDs recalibrated to weeks 68 to 80.

Loss avoided	LR	LR+	CB	CB+
Rejections	3946	3946	3946	3946
True Positive Rejections	586	671	711	814
Loss avoided (millions)	21.9	25.6	28.7	33.0
Fraction of total loss	0.234	0.274	0.307	0.352
Good loans rejected (millions)	229	219	227	217

Table 3: Amount of loss avoided by rejecting the riskiest 3% of applicants using the new models with and without credit bureau data.

Net benefit	LR	LR+	Δ LR	CB	CB+	Δ CB
Net benefit (millions)	3.62	8.14	4.52	10.60	15.63	5.04

Table 4: Net benefit for each of the four models computed by loss avoided minus $0.08\times$ good loans rejected. Columns with Δ indicate the difference in net benefit from the inclusion of the credit bureau data.

ploy the expected loss (EL) formula

$$\text{EL} = \text{PD} \times \text{LGD} \times \text{EAD} \tag{3}$$

where LGD is loss given default, EAD is exposure at default, and PD is the probability of default. We use the variable credamount_770A as the EAD, and we assume that the average LGD is half the amount of the exposure. By rejecting the riskiest 3% of applicants as determined by the fitted PD for each model, we can use the observed default rates to compute the empirical loss as well as the volume of good loans that would be turned away if we reject loans based on our new models with and without credit bureau data. We remark that adjusting the intercept has no effect on this metric, which only relies on the relative ranking of potential borrowers. We report our findings in table 3. The test weeks hold 131,548 cases with 2,788 defaults, 7.38 billion of lending and a total loss of 93.7 million at LGD 0.5.

Similarly, in table 4, we report the net benefit from the inclusion of the credit bureau data for each model given an average profit of 8% on non-defaulting loans. We find that over this time period the credit bureau data increases net benefit by $4.5 million and $5.0 million for logistic regression and CatBoost respectively. As a share of the total lending in the test weeks, the gain is 0.061% for logistic regression and 0.068% for CatBoost, or $34 and $38 per application, which is an approximate upper bound on the most a bank should pay per bureau enquiry; however, since the lender would also pay for enquiries on applicants it declines, the true break-even price is lower.

To check that these differences are not noise, we resample the test cases with replacement 1,000 times and recompute the AUCs and the net benefits at 3% rejection, without refitting the models. Table 5 gives the 95% intervals, and all differences are clearly above or below zero. Since the models are not refitted and the test weeks all fall in the moratorium, these intervals understate the true uncertainty.

95% intervals	AUC	Normalized net benefit (%)
ΔLR	0.031 (0.027, 0.036)	0.061 (0.034, 0.086)
ΔCB	0.026 (0.023, 0.031)	0.068 (0.044, 0.092)
CB $-$ LR	0.019 (0.015, 0.021)	0.095 (0.067, 0.121)
CB+ $-$ LR+	0.014 (0.011, 0.016)	0.102 (0.074, 0.126)

Table 5: Differences in model performance metrics with 95% bootstrap intervals in parentheses. Δ indicates the difference in model performance with and without credit bureau data.

CatBoost without the bureau data has a higher net benefit (10.60 million) than logistic regression with it (8.14 million), even though its AUC is lower. It is important to remark that this amount does not represent the true difference in net profit, as we have no way to account for applicants that were denied and would have been accepted, since our data set is only comprised of approvals.

4 Conclusion

In this report, we examined the impact of external credit bureau data on a logistic regression and gradient boosted decision tree model for personal finance loans around the time of the COVID-19 pandemic. Our results indicate that the inclusion of credit bureau data yields a modest but significant increase in model AUC for both logistic regression and CatBoost, and consequently a small improvement in performance as measured by loss avoided, though we must keep in mind that this value is only one part of the total difference in net profit, as we have not studied the impact of credit data on approvals that would have otherwise been denied. Under the assumptions above, the credit bureau data has a positive value of about \$34 to \$38 per application, and the sign of this effect appears to be robust under bootstrapped resampling. Further research benchmarking these results against non-COVID time periods is necessary to precisely quantify the interaction between shocks and external data sources.

References

[1] N. Siddiqi, *Credit Risk Scorecards: Developing and Implementing Intelligent Credit Scoring*, John Wiley & Sons, Hoboken, NJ, 2006.